Converting Other Equations into Bessel's Form

- Many equations that do not look like Bessel's equation can be transformed into it by a change of the independent and dependent variable.
- **Theorem (Simmons - Book).** If $u(t)$ satisfies Bessel's equation of order n, then for constants a, b, c, the function

$$y(x) = x^a\, u(bx^c)$$

satisfies

$$\boxed{y'' + \frac{1-2a}{x}y' + \left[b^2c^2x^{2c-2} + \frac{a^2 - n^2c^2}{x^2}\right] y = 0}$$

- So the general solution of *this* equation is

$$y = x^a\left[AJ_n(bx^c) + BJ_{-n}(bx^c)\right] \qquad (\text{or } BY_n(bx^c) \text{ if } n \in \mathbb{Z}).$$

- Given a target equation, we match coefficients against this template to read off a, b, c, n.



Worked Illustration: Airy's Equation

- Consider **Airy's equation** $y'' + xy = 0$ (no y' term, and a y-coefficient of x^1).
- Matching against the template: no y' term $\Rightarrow \dfrac{1-2a}{x} = 0 \Rightarrow a = \dfrac{1}{2}$.
- The y-coefficient must equal x, with no $1/x^2$ term present:

$$b^2c^2x^{2c-2} = x^1 \;\Rightarrow\; 2c-2 = 1 \;\Rightarrow\; c = \tfrac{3}{2},\; a^2 - n^2c^2 = 0 \;\Rightarrow\; n = \frac{a}{c} = \frac{1}{3}.$$

- Then $b^2c^2 = 1 \Rightarrow b^2 = \dfrac{1}{c^2} = \dfrac{4}{9} \Rightarrow b = \dfrac{2}{3}$.
- So the general solution of Airy's equation is

$$\boxed{y = \sqrt{x}\left[A\,J_{1/3}\left(\tfrac{2}{3}x^{3/2}\right) + B\,J_{-1/3}\left(\tfrac{2}{3}x^{3/2}\right)\right]}$$

- It is a classical result connecting Airy functions to Bessel functions of order $\pm 1/3$.



Problem : Converting a Fourth-Order-Coefficient Equation

- Convert $x^2y'' + xy' + (x^4 - 4)y = 0$ into Bessel's equation, and state the general solution.
- Dividing by x^2: $y'' + \dfrac{1}{x}y' + \left(x^2 - \dfrac{4}{x^2}\right) y = 0.$
- No y'-coefficient change needed beyond matching $\frac{1-2a}{x} = \frac{1}{x} \Rightarrow a = 0.$
- Matching the y-coefficient: x^2 (no $1/x^2$ term besides $-4/x^2$) gives $2c - 2 = 2 \Rightarrow c = 2$; $b^2c^2 = 1 \Rightarrow b = \frac{1}{2}$; and $a^2 - n^2c^2 = -4$ with $a = 0$ gives $4n^2 = 4 \Rightarrow n = 1$.
- So the general solution is

$$y = A\, J_1\left(\tfrac{1}{2}x^2\right) + B\, Y_1\left(\tfrac{1}{2}x^2\right)$$

(using Y_1, not J_{-1}, since $n = 1$ is an integer).



Parametric Bessel Equation

If $J_\nu(ax)$ and $Y_\nu(ax)$ are solutions, the corresponding equation is the **parametric Bessel equation** with parameter a:

$$x^2y'' + xy' + \left(a^2x^2 - \nu^2\right) y = 0.$$

Reason: Substituting $t = ax$ into the standard Bessel equation, we have to change $\frac{dy}{dx} = a\frac{dy}{dt}$ and $\frac{d^2y}{dx^2} = a^2\frac{d^2y}{dt^2}$ so that

$$t^2\frac{d^2y}{dt^2} + t\frac{dy}{dt} + (t^2 - \nu^2)y = 0.$$

Its general solution is $y(t) = c_1J_\nu(t) + c_2Y_\nu(t)$.
Thus, the general solution of the parametric BDE is

$$y(x) = c_1J_\nu(ax) + c_2Y_\nu(ax).$$



Method 1: Direct Substitution Method

- Find the complete solution of

$$xy'' - y' + 4x^6 y = 0$$

by reducing it to Bessel's form.

- **Method**: Use the substitution

$$y = x^m u(z), \qquad z = ax^p,$$

choosing the constants m, p, a (and the resulting Bessel order n) so that the equation for $u(z)$ becomes exactly Bessel's equation.

- Multiply the given equation by x:

$$x^2 y'' - xy' + 4x^7 y = 0.$$

- This is the standard form required during substitution.



Derivatives under $y = x^m u(z)$, $z = ax^p$

- Since $u = u(z)$ with $z = ax^p$, we have $\dfrac{dz}{dx} = apx^{p-1}$.
- **First derivative**:

$$y' = mx^{m-1}u + x^m u' \cdot \frac{dz}{dx} = mx^{m-1}u + apx^{m+p-1}u'.$$

- **Second derivative** Differentiating again with product rule and chain rule,

$$y'' = m(m-1)x^{m-2}u + ap(2m+p-1)x^{m+p-2}u' + a^2p^2x^{m+2p-2}u''.$$

- Compute

$$x^2y'' = m(m-1)x^m u + ap(2m+p-1)x^{m+p}u' + a^2p^2x^{m+2p}u'',$$

$$xy' = mx^m u + apx^{m+p}u'.$$

- Combining and dividing throughout by x^m:

$$a^2p^2x^{2p}u'' + ap(2m+p-2)x^p u' + \left[m(m-2) + 4x^7\right]u = 0.$$



Change Variable to $z = ax^p$

- Since $x^p = z/a$, substitute directly:

$$p^2 z^2 u'' + p(2m + p - 2)zu' + \left[m(m-2) + 4x^7\right]u = 0.$$

- Dividing by p^2:

$$z^2 u'' + \frac{2m + p - 2}{p} zu' + \frac{m(m-2) + 4x^7}{p^2} u = 0.$$

- We now compare this against **Bessel's equation** of order n:

$$z^2 u'' + zu' + (z^2 - \nu^2)u = 0.$$

- **Coefficient of** zu' must equal 1:

$$\frac{2m + p - 2}{p} = 1 \quad \Longrightarrow \quad 2m = 2 \quad \Longrightarrow \quad m = 1.$$

- **Coefficient of** u: with $m = 1$, $m(m-2) = -1$, so we need

$$\frac{-1 + 4x^7}{p^2} = z^2 - \nu^2 = a^2 x^{2p} - \nu^2.$$



Finding p, a, n

- Matching the powers of x in $\dfrac{4x^7}{p^2} = a^2x^{2p}$:

$$2p = 7 \implies p = \frac{7}{2}, \qquad \frac{4}{p^2} = a^2 \implies a^2 = \frac{16}{49} \implies a = \frac{4}{7}.$$

- Matching the constant terms in $-\dfrac{1}{p^2} = -\nu^2$:

$$\nu^2 = \frac{4}{49} \implies \nu = \frac{2}{7}.$$

- With $m = 1$, $p = \dfrac{7}{2}$, $a = \dfrac{4}{7}$, $\nu = \dfrac{2}{7}$, the equation for u is exactly Bessel's equation of order $\nu = 2/7$:

$$z^2u'' + zu' + \left(z^2 - \frac{4}{49}\right)u = 0, \qquad z = \frac{4}{7}x^{7/2}.$$

- Since $\nu = \dfrac{2}{7}$ is **not an integer**, $J_{2/7}(z)$ and $J_{-2/7}(z)$ are linearly independent solutions:



The General Solution

- Recall $y = x^m u = xu$. Substituting $u(z)$:

$$\boxed{y(x) = x\left[C_1\, J_{2/7}\left(\frac{4}{7}x^{7/2}\right) + C_2\, J_{-2/7}\left(\frac{4}{7}x^{7/2}\right)\right]}$$

- This is the **complete general solution** for $x > 0$, with C_1, C_2 arbitrary constants.
- The same process can be done by using the change of dependent variable first, and then change of independent variable in the next.



Method-2: Two-Stage Strategy

- Find the complete solution of the same problem

$$xy'' - y' + 4x^6 y = 0$$

  by reducing it to Bessel's equation.
- **Stage 1**: substitute $y = x^m u(x)$ (with u still a function of x) and choose m so that the transformed equation has the same zu'-type coefficient structure as Bessel's equation.
- **Stage 2**: substitute $z = ax^p$ (so $u(x) = w(z)$) and choose a, p so that the equation for $w(z)$ becomes *exactly* Bessel's equation, reading off the order n.



Stage-1: Derivatives under $y = x^m u(x)$

- First multiply the given equation by x:

$$x^2 y'' - xy' + 4x^7 y = 0.$$

- With $y = x^m u(x)$:

$$y' = mx^{m-1} u + x^m u',$$

$$y'' = m(m-1)x^{m-2} u + 2mx^{m-1} u' + x^m u''.$$

- So $x^2 y'' - xy' + 4x^7 y = 0$ becomes

$$x^{m+2} u'' + (2m-1)x^{m+1} u' + \left[m(m-2)x^m + 4x^{m+7}\right] u = 0.$$

- Dividing throughout by x^m:

$$x^2 u'' + (2m-1)xu' + \left[m(m-2) + 4x^7\right] u = 0. \qquad (*)$$



Stage-1: Determining m

- Bessel's equation, $z^2w'' + zw' + (z^2 - \nu^2)w = 0$, has coefficient **exactly** 1 on the first-derivative term zw'.

$$2m - 1 = 1 \quad \Longrightarrow \quad \boxed{m = 1},$$

- Substituting $m = 1$ into $(*)$: $m(m-2) = 1(-1) = -1$, so

$$x^2u'' + xu' + \left(-1 + 4x^7\right)u = 0. \tag{I}$$

- This is now in the right "shape" to apply the second substitution $z = ax^p$.



Stage 2: The General Effect of $z = ax^p$

- Write $u(x) = w(z)$, $z = ax^p$, so $\dfrac{dz}{dx} = apx^{p-1}$. A short calculation (chain rule twice) shows that for *any* equation of the form $x^2u'' + xu' + f(x)u = 0$ (i.e. coefficient 1 on xu', as we now have),

$$x^2u'' + xu' \longrightarrow p^2z^2w'' + p^2zw'.$$

- Dividing by p^2:

$$z^2w'' + zw' + \frac{f(x)}{p^2}\,w = 0.$$

- The zw' coefficient is automatically 1, **for any** p – confirming Stage 1's choice of m was the right one to make first.



Stage 2: Applying to Equation (I) for a and p

- Here $f(x) = -1 + 4x^7$, so

$$z^2 w'' + z w' + \frac{-1 + 4x^7}{p^2} w = 0.$$

- We need this to match Bessel's equation exactly:

$$z^2 w'' + z w' + (z^2 - \nu^2) w = 0,$$

i.e.

$$\frac{-1 + 4x^7}{p^2} = z^2 - \nu^2 = a^2 x^{2p} - \nu^2.$$

- Matching the x^7 (i.e. x^{2p}) terms:

$$2p = 7 \implies \boxed{p = \frac{7}{2}}, \qquad \frac{4}{p^2} = a^2 \implies a^2 = \frac{16}{49} \implies \boxed{a = \frac{4}{7}}.$$

- Matching the constant terms:

$$-\frac{1}{p^2} = -\nu^2 \implies \nu^2 = \frac{4}{49} \implies \boxed{\nu = \frac{2}{7}}.$$



The Reduced Bessel Equation

- With $m = 1$, $p = \frac{7}{2}$, $a = \frac{4}{7}$, the equation for $w(z)$ is exactly Bessel's equation of order $n = \frac{2}{7}$:

$$z^2 w'' + z w' + \left(z^2 - \frac{4}{49}\right) w = 0, \qquad z = \frac{4}{7} x^{7/2}.$$

- Since $\nu = \frac{2}{7}$ is **not an integer**, $J_{2/7}(z)$ and $J_{-2/7}(z)$ are linearly independent:

$$w(z) = C_1 J_{2/7}(z) + C_2 J_{-2/7}(z).$$

- Recall $y = x^m u = x \cdot u(x) = x \cdot w(z)$. Substituting:

$$\boxed{y(x) = x \left[C_1 \, J_{2/7}\left(\frac{4}{7} x^{7/2}\right) + C_2 \, J_{-2/7}\left(\frac{4}{7} x^{7/2}\right) \right]}$$

- This is the **complete general solution** for $x > 0$, with C_1, C_2 arbitrary constants, which obtained earlier.



Problem

- Find the complete solution of the differential equation by reducing it to Bessel's form:

$$x^2y'' - 5xy' + 9(x^6 - 8)y = 0.$$

- **Solution** Put $y = x^m u$, where $u = u(x)$.
- Then $y' = mx^{m-1}u + x^m u'$,
  $y'' = m(m-1)x^{m-2}u + 2mx^{m-1}u' + x^m u''$.
- Substituting into the given equation:

$$x^2\left[m(m-1)x^{m-2}u + 2mx^{m-1}u' + x^m u''\right]$$

$$-5x\left[mx^{m-1}u + x^m u'\right] + 9(x^6 - 8)x^m u = 0.$$

- Cancelling x^m throughout the equation, we obtain
- $x^2u'' + (2m-5)x\,u' + \left(m^2 - m - 5m + 9x^6 - 72\right)u = 0.$



Solution: Stage -1

- Comparing the coefficient of xu' with Bessel's equation,

$$x^2u'' + xu' + (x^2 - \nu^2)u = 0,$$

we require

$$2m - 5 = 1, \qquad m = 3.$$

- Thus $y = x^3u$. With $m = 3$, the diff. eq. reduces to

$$x^2u'' + xu' + (9x^2 - 81)u = 0.$$

- Put $z = ax^p$. Then $\frac{du}{dx} = \frac{du}{dz}\frac{dz}{dx} = apx^{p-1}\frac{du}{dz}$.
- $\frac{d^2u}{d^2x} = ap(p-1)x^{p-2}\frac{du}{dz} + apx^{p-1}\frac{d^2u}{dz^2}\frac{dz}{dx} = ap(p-1)x^{p-2}\frac{du}{dz} + a^2p^2x^{2p-2}\frac{d^2u}{dz^2}$.
- Using these computations in the above reduced equation, we have $p(p-1)x^p\frac{du}{dz} + p^2x^{2p}\frac{d^2u}{dz^2} + px^p\frac{du}{dz} + 9(x^6 - 9)u = 0$.
- It is equal to $p^2z^2\frac{d^2u}{dz^2} + p^2z\frac{du}{dz} + 9((\frac{z}{a})^{6/p} - 9)u = 0$, i.e. $z^2\frac{d^2u}{dz^2} + z\frac{du}{dz} + \frac{9}{p^2}((\frac{z}{a})^{6/p} - 9)u = 0$.



Solution: Stage- 2

- Comparing the coefficient of u with Bessel's equation, we need $\frac{6}{p} = 2 \implies p = 3$.
- Then $a^2 = 1 \implies a = 1$ and $\nu = 3$. Thus, $z = x^3$.
- Otherwise, using $\frac{9x^6 - 81}{p^2} = a^2 x^{2p} - \nu^2$, we will get $2p = 6, \quad \frac{9}{p^2} = a^2 \implies a^2 = 1$ and $\frac{81}{p^2} = \nu^2 \implies \nu^2 = 9$.
- The BDE is in this case $z^2 u'' + zu' + (z^2 - 9)u = 0$.
- Since $\nu = 3$ (integer), we have $u(z) = C_1 J_3(z) + C_2 Y_3(z)$.
- Using $y = x^m u = x^3 u(z)$, the general solution is

$$\boxed{y(x) = x^3 \left[C_1 J_3(x^3) + C_2 Y_3(x^3) \right]}$$

.



Problem

- Find the complete solution of $xy'' - y' + 4x^5 y = 0$ by reducing it to Bessel's form:
- **Solution.** Put $y = x^m u$ in the given diff. equation.
- Then the transformed equation is

$$x^2 u'' + (2m-1)xu' + \left(4x^6 + m^2 - 2m\right) u = 0.$$

- Comparing with the coefficient of xu' with BDE, $2m - 1 = 1 \implies m = 1$.
- Hence

$$x^2 u'' + xu' + (4x^6 - 1)u = 0.$$

- Put $z = ax^p$. Then $\frac{du}{dx} = \frac{du}{dz}\frac{dz}{dx} = apx^{p-1}\frac{du}{dz}$.
- $\frac{d^2u}{d^2x} = ap(p-1)x^{p-2}\frac{du}{dz} + apx^{p-1}\frac{d^2u}{dz^2}\frac{dz}{dx} = ap(p-1)x^{p-2}\frac{du}{dz} + a^2p^2x^{2p-2}\frac{d^2u}{dz^2}$.
- Using these computations in the above reduced equation, we have $p^2z^2\frac{d^2u}{dz^2} + p^2z\frac{du}{dz} + [4(\frac{z}{a})^{6/p} - 1]u = 0$.



Solution

- Dividing by p^2, we obtain

$$z^2\frac{d^2u}{dz^2} + z\frac{du}{dz} + \left[\frac{4}{p^2}\left(\frac{z}{a}\right)^{6/p} - \frac{1}{p^2}\right]u = 0.$$

- Now, we need $6/p = 2 \implies p = 3$.
- In this case, we have $\frac{4}{9a^2} = 1 \implies a = \frac{2}{3}$
- Using the values a and p in the above equation, we have BDE

$$z^2\frac{d^2u}{dz^2} + z\frac{du}{dz} + \left[z^2 - \left(\frac{1}{3}\right)^2\right]u = 0.$$

- The solution of BDE with $\nu = \frac{1}{3}$ is given by

$$u = C_1 J_{1/3}(z) + C_2 J_{-1/3}(z) = C_1 J_{1/3}\left(\frac{2}{3}x^3\right) + C_2 J_{-1/3}\left(\frac{2}{3}x^3\right)$$

.

- Thus,

$$\boxed{y(x) = x\left[C_1 J_{1/3}\left(\frac{2}{3}x^3\right) + C_2 J_{-1/3}\left(\frac{2}{3}x^3\right)\right].}$$



Problem

- Find the complete solution of $y'' + y = 0$ by reducing it to Bessel's form:
- **Solution.** Put $y = x^m u$ in the given diff. equation.
- Then
$$x^2 u'' + 2mxu' + \left(x^2 + m(m-1)\right) u = 0.$$
- Comparing with Bessel's equation gives
$$2m = 1, \qquad m = \frac{1}{2}.$$
- Thus
$$x^2 u'' + xu' + \left(x^2 - \frac{1}{4}\right) u = 0.$$
- Therefore $\nu^2 = \frac{1}{4} \implies \nu = \pm\frac{1}{2}$, and
$$u = C_1 J_{1/2}(x) + C_2 Y_{1/2}(x),$$
and
$$\boxed{y(x) = x^{1/2}\left[C_1 J_{1/2}(x) + C_2 Y_{1/2}(x)\right].}$$



Normal Form of Second Order Homogeneous Linear ODE

- Consider a second order linear ODE in the standard form

$$y'' + p(x)y' + q(x)y = 0. \tag{15}$$

- By a change of dependent variable, (15) can be written as

$$u'' + Q(x)u = 0, \tag{16}$$

  which is called the **normal form** of (15).
- To find the transformation, put $y(x) = u(x)v(x)$.
  Substituting into (15) gives

$$vu'' + (2v' + pv)u' + (v'' + pv' + qv)u = 0.$$

- Setting the coefficient of u' to zero:

$$2v' + pv = 0 \implies v = e^{-\int p/2\, dx}.$$



Normal Form of Bessel's Equation

- The coefficient of u becomes

$$\left(q(x) - \tfrac{1}{4}p^2 - \tfrac{1}{2}p'\right) v = Q(x)v.$$

- Since v is nonzero, cancelling v gives the required normal form.
- Since v never vanishes, u vanishes if and only if y vanishes – the transformation has no effect on the zeros of the solution.
- Consider the Bessel equation of order $\nu \geq 0$:

$$x^2y'' + xy' + (x^2 - \nu^2)y = 0, \qquad x > 0.$$

- Here $p = 1/x$, $q = 1 - \nu^2/x^2$, $v = e^{-\int 1/2x\, dx} = 1/\sqrt{x}$. Now

$$Q(x) = 1 - \frac{\nu^2}{x^2} - \frac{1}{4x^2} + \frac{1}{2x^2} = 1 + \frac{1/4 - \nu^2}{x^2}.$$

- Thus, the Bessel equation in normal form becomes

$$u'' + \left(1 + \frac{1/4 - \nu^2}{x^2}\right) u = 0. \tag{17}$$

